Supplemental Material for "Physics-Informed Neural Representation of a Timelike FLRW Geodesic: Conformal First Integrals and Affine Reconstruction"

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I. CARTESIAN EXTENSION TO TWO AND THREE SPATIAL DIMENSIONS

The one-direction sector is sufficient for the main numerical comparison. The same first-integral construction extends to the spatially flat Cartesian metric

$$\mathrm{d}s^2 = a(\eta)^2 \left(\mathrm{d}\eta^2 - \delta_{ij}\,\mathrm{d}x^i \mathrm{d}x^j\right), \qquad i,j = 1,\ldots,d. \tag{1}$$

Spatial translations give the conserved momentum vector

$$P_i = a^2 \frac{\mathrm{d}x^i}{\mathrm{d}\lambda}, \qquad P^2 = \delta^{ij} P_i P_j, \tag{2}$$

and the tangent normalization is

$$a^2 \left[\left(\frac{\mathrm{d}\eta}{\mathrm{d}\lambda}\right)^2 - \delta_{ij}\frac{\mathrm{d}x^i}{\mathrm{d}\lambda}\frac{\mathrm{d}x^j}{\mathrm{d}\lambda}\right] = \kappa. \tag{3}$$

On the future-directed branch,

$$\frac{\mathrm{d}x^i}{\mathrm{d}\eta} = \frac{P_i}{\sqrt{P^2 + \kappa a^2}}, \qquad \frac{\mathrm{d}\lambda}{\mathrm{d}\eta} = \frac{a^2}{\sqrt{P^2 + \kappa a^2}}. \tag{4}$$

Thus $\mathbf{x}(\eta) - \mathbf{x}_0$ remains parallel to the fixed vector $\mathbf{P}$. Increasing the number of Cartesian outputs tests whether the implementation preserves that structure, but it does not introduce new geodesic degrees of freedom. A genuinely coupled extension would require, for example, spatial curvature, a background without these simple translation integrals, or additional fields or inferred geometry.

For the supplementary runs we use the same five-layer, width-64 tanh trunk as in the main calculation, seed 42, 514 final collocation points, and hard initial value/slope constraints. The conformal training interval is $0.5 \le \eta \le 1.5$. The momentum vectors have $P = 1$: $P_i = (1)$, $(0.8, 0.6)$, and $(0.60, 0.48, 0.64)$ for $d = 1, 2, 3$. Each run uses 1250 Adam updates followed by an L-BFGS cap of 120 iterations. PyTorch recorded one realized L-BFGS iteration and two function evaluations in all three runs, so the cap is not a measure of completed quasi-Newton work.

TABLE I. Supplementary fixed-grid conformal extension runs. The vector trajectory error is computed from all Cartesian spatial components on the common affine grid. These runs are an implementation and dimensional-scaling check rather than a new ablation against the affine baselines.

d	$E_{\mathrm{rel}}[\eta(\lambda)]$	$E_{\mathrm{rel}}[\mathbf{x}(\lambda)]$	$\max \lVert\Delta\mathbf{P}\rVert$	$\max \lvert\Delta\kappa\rvert$	$\max \lVert\mathbf{x}_\perp\rVert$
1	1.4744×10^{-4}	2.7513×10^{-4}	1.3304×10^{-2}	5.2879×10^{-2}	0
2	1.3044×10^{-4}	5.7659×10^{-4}	1.1049×10^{-2}	7.5589×10^{-2}	8.16×10^{-5}
3	1.1858×10^{-4}	1.1099×10^{-3}	1.0179×10^{-2}	1.1253×10^{-1}	6.96×10^{-5}

The affine reconstruction gives an η error of order 10^{-4} in all three runs, while the spatial-vector error rises from 2.75×10^{-4} at $d = 1$ to 1.11×10^{-3} at $d = 3$. The transverse components remain below 8.2×10^{-5} in the chosen units. The derivative-sensitive tangent-normalization defect reaches about 0.113 in the three-dimensional run. These runs therefore check the implementation across output dimension without introducing a new dynamical system.

II. REPRODUCTION AND AUDIT PROTOCOL

The main paper reports metrics computed from saved checkpoints and validation arrays. A reproduction of those results is distinct from a new training run, since optimization trajectories can change across software or hardware environments even when the nominal seed and settings are the same.

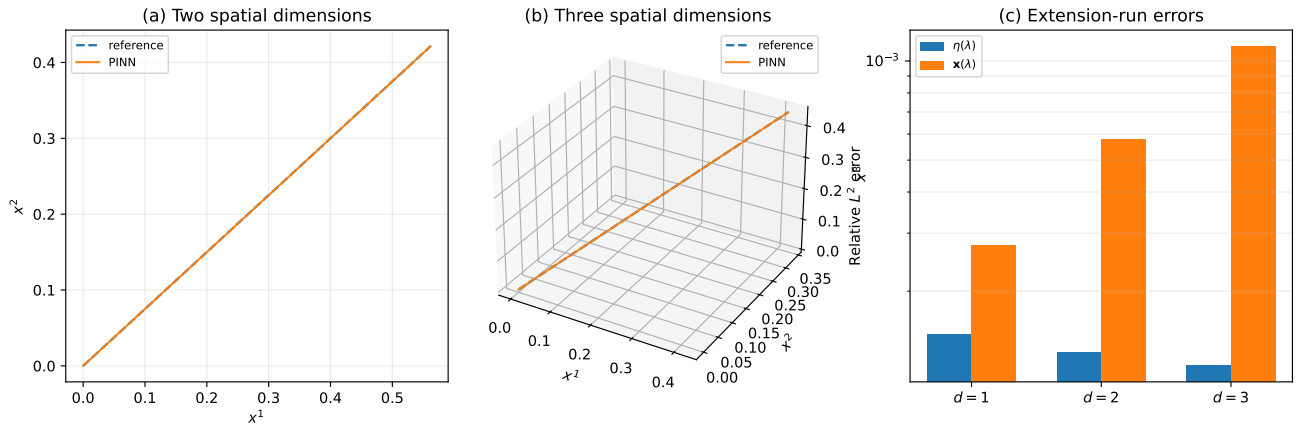


FIG. 1. Supplementary Cartesian dimensional-extension runs. The learned two- and three-dimensional comoving paths follow the reference directions set by $\mathbf{P} = (0.8, 0.6)$ and $\mathbf{P} = (0.60, 0.48, 0.64)$, respectively. The common-affine-grid errors for $d = 1, 2, 3$ are shown in the final panel. Since $P = 1$ in all three cases, the exact $\eta(\lambda)$ and radial pacing are the same.

a. Baseline-result reproduction. The baseline reproduction uses the seed-42 six-cell checkpoint tree. For each cell, the corresponding directory contains the checkpoint, configuration, training history, stage record, and validation arrays as a single provenance unit. The reported trajectory metrics are computed from the saved arrays on the common reporting interval, not from a newly trained model. The post-training audit evaluates each loaded checkpoint on the denser grid without changing its parameters.

b. Reference and inversion sequence. The reference calculation consists of an affine second-order integration with the stated initial data and tolerances and an independent integration of the reduced conformal first-integral equations. The reference $x(\eta)$ and $\lambda(\eta)$ are compared on a common conformal grid before inversion. The learned $\widehat{\lambda}(\eta)$ is tested in its original increasing-η order for derivative sign and endpoint coverage; only then is the monotone PCHIP inverse constructed. Extrapolation is disabled, and direct conformal-coordinate errors are reported separately from errors after inversion.

c. Physical checks. The audit grid tests $\eta > 0$, future direction, tangent norm, and conserved momentum for the affine models. For the conformal models, the corresponding checks use the signed pacing derivative, its increments, and reconstructed invariant defects. These are finite-grid diagnostics: a dense grid can reveal narrow failures, but it does not establish a continuum bound without an additional derivative or interval argument.

d. Supplementary training runs. A supplementary run counts as a direct reproduction only if it preserves the recorded architecture, data type, initialization, collocation schedule, optimizer stages, and stopping settings. Its record includes software versions, hardware, wall-clock scope, realized L-BFGS iteration and function-evaluation counts, and any failed or restarted cell. Robustness claims should use multiple independent seeds and report the distribution of errors rather than a selected run.

III. REPOSITORY MAP

The project archive is organized as follows:

```
flrw-geodesic-pinn/
|-- src/flrw_geodesic_pinn/        reusable physics/ML utilities
|-- scripts/                       reproduction, audit, and figure CLIs
|-- configs/                       machine-readable experiment settings
|-- notebooks/                     reproducibility and archived notebooks
|-- results/                       saved numerical records and controls
|-- tests/                         lightweight source-level checks
|-- docs/                          reproducibility and release notes
|-- CITATION.cff
|-- Makefile
|-- pyproject.toml
`-- requirements.txt
```

The saved seed-42 checkpoints are kept under `results/archived_seed42/`. The original full-budget AFI control is under `results/afi_full_budget/`; the 220-iteration source state used for the continuation is retained under `results/afi_full_budget_incremental/`, and the 500-iteration continuation is under `results/afi_full_budget_500/`. The five-seed padded-CFI screen is under `results/seed_sweep_adam_padded_cfi/`. The dimensional extension is under `results/dimensional_extension/`. Historical notebooks are retained separately from the scripts used for the current calculations.